

RANIA HAAS

CINCINNATI, OH • (419) 261-3160 • haasr3@mail.uc.edu

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Cincinnati

B.S. Computer Science | Certification in Japanese Language and Culture

Graduating: May 2027

GPA: 3.0

SKILLS & CERTIFICATIONS

Programming: C++, C/AL, Svelte, HTML, JavaScript, Python, SQL, CSS, LabVIEW

Software & IDE: DynamicsNAV, Visual Studio Code, Arduino, PlatformIO, KiCAD, Fusion 360, Ubuntu

Certifications: JLPT N4, National Association of Rocketry (NAR) L1

PROJECTS

TOBI – Telemetry, Orientation and Barometric Instrument Flight Computer

May 2026 - Present

- Designed and developed a custom flight computer using an Sseeed ESP32-S3 and Adafruit LSM6DSO32 to monitor live data, prepare system for flight, log and record events, and allow the user to download and access system data through a front-end server using C++ and HTML with FreeRTOS
- Refined the design using a custom PCB created in KiCAD as well as a custom 3D printed electronics bay for flight

Harty Darty and Double Trouble – Custom Avionics Development

August 2025 - Present

- Integrated code with microcontrollers, barometers, and pyrotechnics to create a custom avionics bay for Level 2 drag-separated and two stage rockets with dual redundancy through multi-threading & C++
- Advanced the code and software design by creating embedded software for launch, burnout, apogee, and descent detection, including robust data acquisition, multi-rate filtering, and structured CSV telemetry output for analysis

NAR Level 2 Certification Rocket

Sept 2025 - Present

- Manufactured a 4" diameter 68" tall model rocket intended for a Level 2 (L2) certification utilizing OpenRocket to determine safety margins, create custom 3D printed parts for dual deployment to achieve an apogee of 5,000ft
- Demonstrated advanced knowledge of propulsion, avionics integration, and recovery systems in the mandated L2 test

User Interfaces Portfolio

Aug 2025 – Dec 2025

- Developed 4 separate projects highlighting core features of user interfaces based on ideating, sketching, efficiency, and learnability
- Worked on personal journals, technology-based closets, University of Cincinnati's log-in page, and faith-based health alternatives applications in Svelte

EXPERIENCE

McDonald's Cooperate

Mar 2025 - Mar 2026

Mentee | Chicago, Illinois (Remote)

- Worked alongside an assigned software developer mentor to develop career-oriented projects and personal development skills while solving industry related problems
- Participated in-person with their Global Women in Technology Program highlighting the authenticity and knowledge of female leadership positions in cooperate careers

GMI Companies

Aug 2024 - Aug 2025

Information Services Intern | Lebanon, Ohio

- Developed several code units, reports, and tables to synchronize with previous code to speed up cross-referencing in the sales department, processing department, and several other departments across the factory floor
- Assisted in the maintenance and optimization of Microsoft Dynamics NAV, focusing on proper documentation, optimization, and best outcomes for end-users and external recipients

Code Ninjas

July 2021 - Apr 2023

Code Sensei | Toledo, Ohio

- Invented and implemented engaging coding challenges that fosters a hands-on learning environment that boosted students' problem-solving skills and creativity in JavaScript, HTML, and Lua
- Cultivated personal programming and engineering skills by designing and developing custom coding exercises projects for students to help promote software development skills

ACTIVITIES

University of Cincinnati Rocketry Club

July 2025 – Present

Customs Lead, Member, Level 2 Mentorship Program Mentee

- Collaborated as lead and member of custom avionics to develop custom flight computers two-stage deployments
- Worked under the President as a mentee to develop a Level 2 rocket for certification as well as structures skills

Seeking Full-time Positions May 2027